

Global Human Settlements

Overview of twelve Laudato Si analysis maps

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Purpose and deliverables

The new ArcGIS Pro project reconnects global settlement indicators with corrected LSAP members, dioceses and episcopal conferences. It preserves ecclesiastical foundations, reproduces the final prior analyses and adds descriptive aliases, ready-to-use short and long summaries, ranks and percentiles. This overview indexes twelve separate map documents and one source-data document with evidence and an archived workflow.

Project: GHSL/LSAP_Global_Human_Settlements.aprx. Database: GHSL/Global_Human_Settlements.gdb. There are 15 maps: 12 analyses, a CorrectGeo reference map, a source map and a foundation map. The documentation describes the saved state on 4 October 2026.

Map	Analysis	Scientific question
01	Urban expansion and population	Built-up surface growth is compared with population growth over 2015–2025.
02	Green surroundings and urban residents	This is the share of urban residents living in high-green areas in 2025.
03	Heat and green surroundings	The screening combines highest-quarter 2020 heat days and lowest-quarter 2025 high-green residential share among cities containing corrected LSAP members.
04	Urban flood exposure	The 2025 population share refers to areas exposed to a 100-year return-period flood.
05	Groundwater and population growth	The focus combines a positive groundwater-transgression indicator with positive population growth in the global upper quarter, 2015–2025.
06	Urban resilience profile	Growth, heat, flood exposure, and green surroundings are displayed as separate indicators.
07	Environmental exposure and health facility proximity	The focus combines relatively low hospital or pharmacy proximity with relatively high PM2.5, heat, or positive flood exposure among member cities.
08	Education infrastructure and LSAP presence	The 2020 source education infrastructure indicator is retained without interpreting it as a count of schools or a measure of Catholic education.
09	Urban trees, greenness and heat	The principal comparison pairs 2020 canopy height, green coverage, and heat.
10	Recorded multiple-hazard context	The count records how many of eleven hazard-type fields are positive in the 2015 source reference period.
11	Urban cropland and built-up change	The focus is cropland decline in 2015–2019 alongside built-up growth in 2015–2025.
12	Cities within corrected 100 km member reach	Which mapped cities intersect corrected member buffers and how is that reach distributed?

Results at a glance

Map	Global focus	Connected city focus	Classifiable global cities
01	6,880	210	11,422
02	4,508	160	11,420
03	1,538	38	11,421
04	3,466	160	11,422
05	1,209	60	11,422
06	Unavailable	Unavailable	Unavailable
07	1,256	107	4,699
08	Unavailable	Unavailable	Unavailable
09	712	74	11,421
10	10,549	605	11,422
11	326	30	11,422

Connected means one or more CorrectGeo member points intersect the city, not a 100 km reach. The 638 connected cities contain 1,929 matched point records; 810 records have no city match. Maps 06 and 08 have no focus screen. Thresholds and valid denominators vary by metric and screen.

Map 12 reaches 4,027 distinct cities through 26,929 buffer-city relationships; 2,666 of 2,739 member buffers intersect a mapped city. Its jurisdiction summaries exclude cross-boundary cities using unique assignment. These counts answer a different spatial question from direct point-city matching.

Key distribution checks

All 11,422 cities have positive recorded built-up surface change. Cropland decreases in 326 cities, increases in 215 and is unchanged in 10,881. The 10,549 multi-hazard-focused cities have at least two positively recorded hazard types; this does not mean other cities are free of hazards.

Ecclesiastical data

All 43 diocesan and 186 conference original attributes were preserved and compared value-for-value. Names, rites, traditions, hierarchy, source dates, population, Catholic statistics, clergy, parishes, leadership, enrolment, education and duplicate source series remain available. Member and buffer source attributes and geometry also remain. GS_Ecclesial provides readable context in longer summaries. Leadership and historic statistics retain source dates; they were not revalidated as current officeholders. Fields with suffix _26 and other original series were not merged or silently overwritten.

Common spatial and statistical methods

Cities use whole mapped urban centres. Members inherit values only after one unique city match: 1,929 of 2,739 point records match 638 cities; 810 unmatched records keep null indicators and zero city-linked counts. Dioceses, conferences and member buffers use an equal-city mean of distinct intersecting cities, separately for each metric's valid observations. These are neither population-weighted nor area-weighted territorial estimates. A cross-boundary city can occur in multiple summaries, so totals across overlapping territories are not additive.

Null, nonnumeric, nonfinite and sentinel values ≤ -999 are excluded. Negative values are excluded for nonnegative measures, but signed indices and changes can remain negative. Percentages must be within 0–100; heat days within 0–366. Percentage changes require a positive baseline. Compound screening uses three-valued logic: AND is false if any component is false, true if all are true, otherwise unknown; OR is true if any is true, false if all are false, otherwise unknown.

Rank 1 is the highest numeric value, with competition ranks for ties. Percentile is $100 \times (\text{number below} + \text{half the number equal}) / \text{valid N}$. Higher percentile means a higher numeric value, not better or worse conditions. Missing values are excluded. All-equal values have rank 1 and percentile 50.

Five separate metric ranking groups are used: global cities, dioceses, conferences, 20 km buffers and 100 km buffers. Member records inherit global city ranks. The field's RankN gives its valid comparison size. Aggregate focus percentages use classifiable cities as the denominator, not residents or members. Nearest-rank quartiles include ties; map 09 uses interpolated global paired-data terciles.

Map specific screens

Map	Screen or method
01	Focus when built-up percentage growth minus population percentage growth is greater than zero. Growth uses a strictly positive baseline; hectares are converted from square metres before calculating changes.
02	Focus when GreenPop $\leq 1.5619841814041138\%$. This is the nearest-rank lower quartile among 638 connected cities with valid data.
03	Focus when HeatDays ≥ 248 and GreenPop $\leq 1.5619841814041138\%$. Both quartiles use the 638 connected cities with paired valid heat and green observations.
04	Focus when FloodPop > 0 and $\geq 5.079999923706055\%$. The upper quartile is calibrated on connected cities.
05	Focus when Groundwater > 0 and PopGrowth > 0 and $\geq 13.65561811911165\%$. Population growth uses the global upper quartile. GS_Recharge and GS_Salinity retain WA_GWR_MMA_2025 and WA_GWS_SAL_2025 categories separately.
06	The four indicators remain separate. HeatDays is the display and primary ranking field. No combined resilience score or focus classification is calculated.
07	Focus = (Hospital $\leq 18.68016815185547\%$ OR Pharmacy $\leq 7.677029132843018\%$) AND (positive PM25 $\geq 22.209293365478516 \mu\text{g}/\text{m}^3$ OR positive HeatDays ≥ 248 OR positive FloodPop $\geq 5.079999923706055\%$). Each threshold uses its own valid connected-city observations.
08	Retain the CISI education sector source indicator without rescaling it into numbers of schools. No focus classification is calculated.
09	Focus when HeatDays ≥ 259 and GreenCover20 $\leq 0.04457193737228712\%$. Use global paired-valid 2020 observations, with upper two-thirds heat and lower one-third greenness interpolated quantiles. The green threshold is literally 0.04457%, not 4.457%. GreenChange is 2025 minus 2020 coverage in percentage points.
10	Count positive values among FLO, HEW, WLF, DRO, EWI, TCY, LAN, COW, EAR, TSU and VOL. Focus when HazardTypes ≥ 2 . GS_HazardCombo stores names of recorded types; GS_HazardValid counts usable numeric fields, which is not an assessment of reporting coverage.
11	Focus when CropChange < 0 and BuiltChange > 0 . Cropland hectares compare 2015–2019; built-up hectares compare 2015–2025. Percentage cropland change requires a positive 2015 baseline; zero-baseline percentage change remains null.

Popup and sidebar use

Maps 01–11 share six analysis datasets and expose theme-specific fields. GS_Short01 through GS_Short11 contain concise popup text; GS_Long01 through GS_Long11 contain longer selected-feature narratives. GS_Ecclesial supplies readable church context. Map 12 uses GS_Reach100Short and GS_Reach100Long. All original ecclesiastical columns remain available for additional popup sections. The package supplies content for a sidebar; it does not publish an ArcGIS Online application.

GS_X stores a metric, GS_X_Rank its descending rank, GS_X_Pctile its midpoint percentile, GS_X_RankN its valid comparison group and GS_X_CityN its valid city contributions. GS_Focus## is 0/1/null at city and member scale but a percentage at aggregate scale. Preserve descriptive aliases, units and reference years when configuring an application.

Quality control summary

Saved calculations and valid city counts were checked after writing. All 28 metric rankings were independently verified for cities, dioceses, conferences, 20 km buffers and 100 km buffers. Aggregate means were reproduced from saved relationship keys. Original geometry and attributes were compared for the five member and ecclesiastical foundations. This validates the implemented calculations, not the accuracy of every publisher indicator or a new independent spatial-overlay algorithm.

Nine prior final analysis families were compared in their documented fields with zero differences. Maps 02 and 06 reuse independently checked components rather than having a separate prior-map comparison; map 04's prior focus flag and map 09's whole screen were not independently compared. Map 12 now has a new saved-relationship check covering nine fields across all cities, dioceses and conferences with zero differences.

The saved project has no broken layers and 88 popup checks passed. Computational agreement is not external ground truth. The source document and map documents specify the evidence files and the limits of each QC claim.

Science summary

The twelve maps organize urban growth, vegetation, thermal stress, flood exposure, groundwater context, resilience components, environmental exposure and facility proximity, education infrastructure, tree and green characteristics, recorded hazards, cropland change and member reach. They support local ecological discernment and questions for verification. Different reference years, source definitions, city boundaries and unweighted summary choices must accompany any interpretation. Rankings describe numbers and do not by themselves identify better conditions or intervention priorities.

Publisher context: <https://human-settlement.emergency.copernicus.eu/ucdb2024Overview.php> and https://human-settlement.emergency.copernicus.eu/ghs_ucdb_2024.php. The JRC overview describes 11,422 urban centres and integrated thematic indicators. Local exports are labelled R2024A; their precise minor version is not proven by the current website release. Source reference years can include modelled estimates and are not direct contemporary surveys.

Complete work record

The project was built from the inspected original GHSL and CorrectGeo sources and recovered final notebook methods. All foundation fields were retained, stable join keys created, seven spatial relationships built, 28 metrics calculated, thematic screens applied, six context layers summarized and five ranking groups verified. The 100 km reach analysis retained prior unique jurisdiction assignment. Twelve maps were configured with themed symbology, country context, aliases and short and long text. Ecclesiastical summaries were enriched, unmatched-member counts corrected, project connections and prior computations checked, and full exports, field documentation and visual overview produced.

This documentation round adds twelve dedicated map documents, one dedicated source-data document, the consolidated overview, per-map field extracts, source field inventories, QC evidence copies and an archived reproducible workflow. A fresh read-only reach check was added. No analysis values were changed during documentation.

Document navigation

Document	Folder
01_Analysis_Documentation.docx	01 Urban expansion and population Documentation
02_Analysis_Documentation.docx	02 Green surroundings and urban residents Documentation
03_Analysis_Documentation.docx	03 Heat and green surroundings Documentation
04_Analysis_Documentation.docx	04 Urban flood exposure Documentation
05_Analysis_Documentation.docx	05 Groundwater and population growth Documentation
06_Analysis_Documentation.docx	06 Urban resilience profile Documentation
07_Analysis_Documentation.docx	07 Environmental exposure and health facility proximity Documentation
08_Analysis_Documentation.docx	08 Education infrastructure and LSAP presence Documentation
09_Analysis_Documentation.docx	09 Urban trees greenness and heat Documentation
10_Analysis_Documentation.docx	10 Recorded multiple hazard context Documentation
11_Analysis_Documentation.docx	11 Urban cropland and built up change Documentation
12_Analysis_Documentation.docx	12 Cities Within 100 km Reach Documentation
Source_Data_Documentation.docx	13 Source Data Documentation

Each analysis folder also contains Theme_Field_Dictionary.csv. The source folder contains the complete Field_Dictionary.csv, exact Source_Dataset_Inventory.csv, Preserved_Source_Fields.csv, original source inventory, Evidence and Workflow. Existing large source and analysis datasets remain in GHSL and are not duplicated here.

Reproduction and dependencies

Open the ArcGIS Pro project and retain the GHSL geodatabase, local CorrectGeo data and country shapefile. This is not a portable GIS package. The evidence and workflow archive are sufficient to inspect how values were produced; rerunning requires the same input sources and ArcGIS Pro with arcpy. SHA256_Manifest.csv records documentation-package file hashes for later comparison.